

NUTRITION MYTHS EXPOSED

The Top 10 Absurdities That Coaches Repeat Every Day (And How Science Debunks Them)

A Science-Based Guide by Adriano

INTRODUCTION

As a nutrition coach who has made countless mistakes and learned from them, I've witnessed firsthand how misinformation spreads like wildfire in our industry. Every day, well-meaning coaches and nutritionists repeat myths that not only lack scientific backing but can actually harm their clients' progress.

This guide exposes the top 10 absurdities I hear daily in the nutrition coaching world, backed by solid science and my own painful learning experiences. From my early days of recommending 48-hour fasts after "cheat days" to understanding the real science behind hunger and satiety, this journey has taught me that good intentions without proper knowledge can be dangerous.

THE TOP 10 NUTRITION ABSURDITIES

1. "DO A 48-HOUR FAST AFTER EATING TOO MUCH"

The Myth: After a binge or overeating episode, coaches recommend extended fasting to "reset" or "compensate" for the excess calories.

Why It's Dangerous: This creates a destructive binge-restrict cycle that actually increases the likelihood of future binges.

The Science: Research by Polivy & Herman (1985) demonstrates that dietary restraint following overeating episodes increases psychological stress and triggers the "what-the-hell

effect," leading to more frequent and severe binges. The restrict-binge cycle is well-documented in eating disorder literature.

My Personal Mistake: I used to train for hours in a fasted state the day after overeating, thinking I could "burn off" the excess calories. I'd wake up after a big meal day and immediately hit the gym for 2-3 hour sessions without eating, believing this would compensate for my previous day's intake. This only made me more likely to overeat again due to extreme hunger and fatigue.

The Solution: Normalize eating patterns immediately. Return to regular meal timing and portions the very next meal. Focus on psychological recovery, not caloric compensation.

2. "CUT ALL CARBOHYDRATES TO LOSE WEIGHT"

The Myth: Carbohydrates are inherently fattening and must be eliminated for weight loss.

Why It's Wrong: Weight loss is determined by caloric deficit, not macronutrient composition. Carbohydrates are not uniquely fattening.

The Science: The DIETFITS study (Gardner et al., 2018) followed 609 adults for 12 months comparing low-fat vs. low-carb diets. Both groups lost similar amounts of weight when calories were controlled. Meta-analyses consistently show that caloric deficit, not carb restriction, drives weight loss (Mansoor et al., 2016).

My Personal Experience: I spent two years avoiding carbs completely, which led to:

- Constant fatigue during CrossFit sessions
- Social isolation (couldn't eat with friends)
- Eventual massive carb binges
- Worse relationship with food

The Solution: Focus on caloric deficit through any sustainable macronutrient distribution. Include carbohydrates for energy, especially if you're active.

3. "DON'T EAT AFTER 6 PM"

The Myth: Eating late at night causes weight gain because metabolism slows down and food is stored as fat.

Why It's Nonsense: Your body doesn't have a magical cutoff time where calories suddenly become more fattening.

The Science: Scheer et al. (2009) found that while circadian rhythms affect metabolism slightly, the effect is minimal (less than 10% variation). What matters is total daily caloric intake, not timing. The "late eating causes weight gain" correlation exists because people who eat late often consume more total calories.

My Personal Experience: I used to avoid eating carbohydrates after 6 PM, thinking they would be stored as fat overnight. I'd eat my last carbs at lunch and then only protein and vegetables for dinner. This led to intense carb cravings at night and eventual late-night binges on bread, pasta, or anything carb-heavy I could find.

The Solution: Eat when it fits your schedule and helps you maintain your caloric goals. If eating later helps you avoid overeating during the day, do it.

4. "DETOX TEAS RESET YOUR METABOLISM"

The Myth: Special teas, cleanses, or detox products can "reset," "boost," or "fix" your metabolism.

Why It's Marketing BS: Your liver and kidneys already detox your body 24/7. No tea can improve this process.

The Science: There's no scientific evidence that any commercial detox product improves metabolic function. Your liver processes toxins continuously and efficiently without external help (Klein & Kiat, 2015). "Metabolic reset" is not a real physiological process.

My Personal Experience: I used to drink green detox juices every morning - kale, pineapple, spirulina, and ginger - believing they were "cleansing" my system and boosting my metabolism. I spent considerable money on spirulina and other superfoods, thinking they had magical properties beyond basic nutrition.

The Solution: Save your money. Focus on adequate sleep, regular exercise, and proper nutrition to support your naturally functioning metabolism.

5. "EAT 6 SMALL MEALS TO BOOST METABOLISM"

The Myth: Eating frequently "stokes the metabolic fire" and burns more calories through increased thermic effect of food (TEF).

Why It's Wrong: Total TEF depends on total food intake, not frequency. Six small meals vs. three larger meals with the same total calories produce identical metabolic effects.

The Science: Bellisle et al. (1997) and Cameron et al. (2010) found no metabolic advantage to increased meal frequency. TEF is proportional to total caloric intake, not meal frequency. Some studies even suggest fewer, larger meals may have slight metabolic advantages.

Personal Reality Check: I was extremely rigid with meal timing, eating every 2-3 hours on the dot, carrying Tupperware everywhere. I set alarms on my phone and would stress if I was even 30 minutes late for a meal, thinking my metabolism would "shut down." It was exhausting, socially awkward, and provided zero additional benefits compared to eating when hungry.

The Solution: Eat in a pattern that fits your lifestyle and helps you maintain your caloric goals. Some people thrive on 2 meals, others on 5. Both work equally well.

6. "NEGATIVE CALORIE FOODS BURN MORE CALORIES THAN THEY CONTAIN"

The Myth: Foods like celery, grapefruit, and cucumber require more energy to digest than they provide, creating "negative calories."

Why It's Physically Impossible: The thermic effect of food is typically 8-15% of calories consumed. No food requires 100%+ of its calories to digest.

The Science: Even the most "thermogenic" foods like protein have a TEF of only 20-30% (Westerterp, 2004). Celery contains about 6 calories per stalk and requires less than 1 calorie

to digest. Basic physics and thermodynamics make negative calorie foods impossible.

The Reality: These foods are low-calorie and high in fiber, making them great for satiety and nutrition, but they're not magical calorie burners.

The Solution: Include low-calorie, nutrient-dense foods for health and satiety, but don't expect them to create caloric deficits through digestion.

7. "CONFUSE YOUR MUSCLES TO PREVENT PLATEAUS"

The Myth: Constantly changing exercises prevents your body from "adapting" and hitting plateaus.

Why It's Counterproductive: Muscle growth requires progressive overload, which is impossible if you're constantly changing exercises.

The Science: Progression in strength and muscle mass requires consistent stimulus with gradually increasing demands (Kraemer & Ratamess, 2004). "Muscle confusion" prevents the adaptation we actually want - getting stronger and building muscle.

My Training Mistake: I used to believe that I needed to feel pain and sweat profusely in every workout to see progress. I thought that if I wasn't completely exhausted and sore the next day, I hadn't worked hard enough. This led to overtraining, burnout, and actually slower progress because I couldn't recover properly between sessions.

The Solution: Stick with basic compound movements for 6-12 weeks, progressively increasing weight, reps, or sets. Change exercises only when progress genuinely stalls.

8. "ARTIFICIAL SWEETENERS CAUSE WEIGHT GAIN"

The Myth: Artificial sweeteners trigger insulin responses, increase cravings, and cause weight gain despite having no calories.

Why It's Misunderstood: Correlation studies are often misinterpreted as causation, and controlled trials show the opposite effect.

The Science: Systematic reviews and meta-analyses (Rogers et al., 2016; Toews et al., 2019) consistently show that artificial sweeteners aid weight loss when replacing sugar. While some observational studies show correlations with weight gain, these are confounded by overall diet quality and lifestyle factors.

Practical Reality: People who drink diet soda often have other unhealthy habits. The diet soda isn't causing weight gain; the overall lifestyle pattern is.

The Solution: Use artificial sweeteners as tools to reduce caloric intake if they help you avoid sugar. Don't fear them based on misinterpreted research.

9. "YOU CAN'T BUILD MUSCLE IN A CALORIC DEFICIT"

The Myth: Muscle building requires a caloric surplus. It's impossible to gain muscle while losing fat.

Why It's Oversimplified: Body recomposition (simultaneous fat loss and muscle gain) is well-documented, especially in certain populations.

The Science: Barakat et al. (2020) meta-analysis found significant muscle gain during caloric deficits in trained individuals when protein intake was adequate (1.6-2.2g/kg body weight). Beginners, detrained individuals, and those with higher body fat percentages show even greater recomposition potential.

The Solution: Prioritize protein intake (1.6-2.2g/kg), maintain resistance training, and don't fear moderate caloric deficits if body recomposition is your goal.

10. "CHEAT MEALS RESET LEPTIN LEVELS"

The Myth: Weekly "cheat meals" or "refeed days" reset the hormone leptin, boosting metabolism and preventing adaptive thermogenesis.

Why It's Oversimplified: Leptin changes require sustained caloric increases over days or weeks, not single meals.

The Science: Leptin levels respond to sustained changes in energy balance over 72+ hours (Chin-Chance et al., 2000). A single high-calorie meal causes temporary leptin increases that return to baseline within 24 hours, providing no metabolic benefit. Strategic refeeds require 2-3 days of increased calories to meaningfully impact leptin.

The Psychological Trap: I used to have designated "cheat days" where I would eat massive amounts of food - an entire pizza, a whole pint of ice cream, or multiple desserts - often without even being hungry or truly wanting the food. I felt like I had to "make the most" of my cheat day, which led to uncomfortable overeating and guilt the next day.

The Solution: If using refeeds, plan 2-3 days of moderate caloric increases (200-300 calories above maintenance) rather than single massive meals. Better yet, take periodic diet breaks of 1-2 weeks at maintenance calories.

THE REAL SCIENCE OF HUNGER AND SATIETY

Understanding hunger isn't just about calories in vs. calories out. Based on current research and the principles outlined in "Desconstruindo o Hábito da Fome," hunger regulation involves complex interactions between:

Hormonal Signals

- **Ghrelin:** The "hunger hormone" that increases before meals
- **Leptin:** The "satiety hormone" that signals fullness
- **CCK, GLP-1, PYY:** Short-term satiety signals from the gut
- **Insulin:** Affects both hunger and fat storage

Environmental Factors

- **Food availability and variety**
- **Social eating cues**
- **Stress and emotional states**

- Sleep quality and duration
- Meal timing and routine

Psychological Components

- Food associations and memories
 - Restriction-induced cravings
 - Emotional eating patterns
 - Body image and self-perception
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MY JOURNEY: FROM MYTH-BELIEVER TO EVIDENCE-BASED COACH

When I started my transformation in 2018, I believed and practiced most of these myths:

- I did 48-hour fasts after weekend binges
- I avoided carbs completely for months
- I carried 6 meals everywhere in Tupperware
- I changed my workout every week
- I spent hundreds on detox teas and supplements

The result? Four years of yo-yo dieting, constant stress about food, and repeated failures.

What changed everything? Discovering I had ADHD and understanding that my brain needed different strategies. I started reading actual research instead of Instagram posts. I learned that sustainable change comes from understanding science, not following trends.

The breakthrough: Realizing that my clients needed education, not restriction. They needed to understand WHY certain approaches work, not just WHAT to do.

PRACTICAL APPLICATIONS FOR COACHES

How to Avoid Spreading Myths:

1. **Question everything** - If it sounds too good to be true, it probably is
2. **Read primary research** - Don't rely on social media interpretations
3. **Understand correlation vs. causation** - Most nutrition myths come from misinterpreted observational studies
4. **Consider individual context** - What works for one person may not work for another
5. **Admit when you don't know** - It's better to say "I don't know" than to guess

Red Flags in Nutrition Information:

- Claims of "resetting" or "boosting" metabolism
 - Promises of rapid results without effort
 - Demonization of entire food groups
 - Testimonials without controlled studies
 - Complex explanations for simple concepts
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CONCLUSION

The nutrition industry is flooded with myths because they're often more appealing than the truth. The truth is that sustainable weight management requires:

- Consistent caloric deficit for weight loss
- Adequate protein for muscle preservation
- Regular resistance training
- Sustainable eating patterns

- Patience and consistency

There are no shortcuts, no magical foods, and no perfect timing. There's only science, consistency, and the courage to admit when we're wrong.

As coaches, our responsibility is to educate based on evidence, not to perpetuate myths because they're popular. Our clients deserve better than recycled Instagram wisdom.

Remember: The best nutrition plan is the one your client can follow consistently for years, not the one that promises miraculous results in weeks.

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This guide represents current scientific understanding as of 2024. Nutrition science evolves, and recommendations may change as new evidence emerges. Always consult with qualified healthcare professionals for personalized advice.

About the Author: Adriano is a certified nutrition coach with advanced training in change psychology from Precision Nutrition. His approach combines evidence-based nutrition science with real-world experience from his own 84-pound transformation and work with diverse clients. He specializes in helping individuals with ADHD and those recovering from

restrictive eating patterns develop sustainable, science-based approaches to nutrition and health.